

EisaX — تقرير الاستثمار الذكي

مدعوم بالذكاء الاصطناعي | تحليل مالي متقدم | للمستثمرين المحترفين

Portfolio uploaded: NVDA, AAPL, MSFT, TSLA, CASH ✓

EisaX Portfolio Risk Report

Date: March 30, 2026 | Period: 1 Year | Risk-Free Rate: 4.5% (US T-Bill)

Executive Summary

Portfolio returned **+29.1%** vs S&P 500 **+14.1%** (Alpha: **+14.9%**). Risk profile: **Aggressive**.
Sharpe: **0.85** (rf=4.5%, 1Y Historical). Immediate action: Reduce tech concentration & add defensive .assets

Holdings

Div Yield	P/E	Beta	Sector	Weight	Ticker
0.02%	34.1	2.38	Technology	30.0%	NVDA
0.41%	31.5	1.12	Technology	20.0%	AAPL
0.95%	22.3	1.11	Technology	20.0%	MSFT
0.00%	332.0	1.93	Consumer Cyclical	15.0%	TSLA
0%	—	0	—	15.0%	CASH

Weighted Portfolio Dividend Yield: 0.33%

Risk Metrics

Method: Historical Simulation (252 trading days) · rf = 4.5%

Assessment	Value	Metric
Strong ●	29.1%+	1Y Total Return
Outperforming benchmark ●	14.9%+	vs S&P 500 (Alpha)
High ●	30.6%	Annualized Volatility
Acceptable ●	0.85	Sharpe Ratio (1Y, rf=4.5%)
Good downside-adjusted ●	1.30	Sortino Ratio (1Y, rf=4.5%)
High Risk ●	1.70	Portfolio Beta (weighted)
of days, loss $\leq$ this 95%	2.41%-	VaR 95% 1-Day (Historical)
Avg loss when VaR is breached — tail risk	3.93%-	CVaR 95% (Expected Shortfall)
Worst peak-to-trough	20.0%-	Max Drawdown (1Y)
Longest time underwater	102d	Max Drawdown Duration
Weighted annual income	0.33%	Portfolio Div Yield
Current (last 63d): -3.38	—	Rolling Sharpe (63d)

Sector Exposure

HHI Contribution	Weight	Sector
0.678	82.4%	Technology
0.031	17.6%	Consumer Cyclical

Sector HHI: 0.709 — ● **Highly Concentrated** (0=perfect diversification, 1=single sector)

Correlation & Diversification

TSLA	MSFT	AAPL	NVDA	
0.57	0.56	0.52	1.00	NVDA
0.52	0.39	1.00	0.52	AAPL

TSLA	MSFT	AAPL	NVDA	
0.43	1.00	0.39	0.56	MSFT
1.00	0.43	0.52	0.57	TSLA

Interpretation	Value	Metric
Moderate co-movement between holdings 🟡	0.50	Avg Pairwise Correlation
Moderate — true independent bets in portfolio 🟡	2.3	Effective N (eigenvalue)

Note: Even without pairs >0.70, Effective N = 2.3 indicates portfolio behaves like **2.3** ⚠️ **independent assets** — not 4. Sector concentration (not just pairwise correlation) is the real risk driver here

Stress Testing 🧪

Historical crises use actual observed returns + sector-adjusted amplification (NOT linear beta)

Notes	Portfolio .Est	SPX Actual	Scenario
Tech -33% vs SPX -19.5% in 2022	28.3%-	19.5%-	Rate Shock (actual) 2022 🛑
SPX -34% in 33 days	40.8%-	34.0%-	COVID Crash Mar 2020 🛑
Financials led; tech -40%	37.8%-	42.0%-	GFC (Sep–Nov) 2008 🛑
Projected; tech typically amplified	39.0%-	30.0%-	Bear Market -30% Scenario 🛑
Tech +55% vs SPX +26.5%	47.7%+	26.5%+	AI Bull Run (actual) 2023 🟢
Growth stocks benefit most	21.0%+	15.0%+	Rate Cut Cycle +15% 🟢

Tech Weight: 100% of equity. In risk-off events, tech typically amplifies market moves by 1.2–1.8×. Linear beta alone ($\beta=1.70$) understates true drawdown risk

Aggressive — $\beta=1.70$, CVaR=-3.93%/day, Eff.N=2.3 bets. 82% in Technology. In a 2022-style selloff ●
your portfolio would have lost ~33% vs SPX -19.5%. Diversification is urgently needed

Performance Attribution (1Y)

Brinson-Hood-Beebower: each holding's contribution to total portfolio return

Attribution	Contribution	1Y Return	Weight	Ticker
●	18.63pp+	52.8%+	35.3%	NVDA
●	6.58pp+	37.3%+	17.6%	TSLA
●	3.45pp+	14.7%+	23.5%	AAPL
●	1.20pp-	5.1%-	23.5%	MSFT
●	1.60pp+	—	—	CASH/Other
	29.06pp+	—	100%	TOTAL

Alpha driver: NVDA contributed +18.6pp (+52.8% × 35% weight). S&P 500 returned +14.1% — 
.alpha gap of +14.9pp

Factor Exposure Analysis

Fama-French proxy: factor tilts computed from fundamentals + price momentum

Interpretation	Portfolio Exposure	Factor
High market sensitivity ●	1.70+	Market (Beta)
Strong growth tilt — high valuation risk ●	0.42+	Growth (P/E tilt)
Large-cap dominated (low SMB) ●	0.82-	Size (SMB proxy)
Strong momentum ●	0.27+	Momentum (12M)

Factor exposures are approximated from fundamentals. For precise Fama-French loadings, a 3-year
.return regression against HML/SMB/MKT factors is required

Institutional Optimization Engine

Constraint Diagnostics 🔍

Reference: \$100,000 portfolio · normalized weights

Required Fix	Status	Current	Limit	Constraint
Sell ~\$15,294 of NVDA	● 15.3pp-	35.3%	20%≥	Single stock: NVDA
Sell ~\$3,529 of AAPL	3.5pp- ●	23.5%	20%≥	Single stock: AAPL
Sell ~\$3,529 of MSFT	3.5pp- ●	23.5%	20%≥	Single stock: MSFT
Reduce by ~\$42,353 / add non-Technology	● 42.4pp-	82.4%	40%≥	Sector: Technology
Add low-β assets: TLT, GLD, BRK-B, XLV	0.40+ ●	1.70	1.30≥	Portfolio beta

Estimated rebalancing: ~\$22,353 turnover (22.4% of portfolio) to reach basic mandate compliance 📊

Mandate Mode Feasibility 📊

Aggressive 🚀	Balanced ⚖️	Conservative 🏛️	
20%	15%	10%	Max single stock
1.50≥	1.20≥	1.00≥	Max portfolio β
40%	35%	30%	Max sector weight
Need 5+ holdings ❌	Need 7+ holdings ❌	Need 10+ ❌ holdings	Single-stock achievable
✅	✅	Min β=1.13 ❌	Beta achievable
Current assets ⚠️ only	Current assets ⚠️ only	Current assets ⚠️ only	Sector cap achievable

:Constraint issues with current holdings ⚠️

- holdings $\times$ 10% = 40% < 100% (need ≥ 10 holdings for this mandate) 4 •
- holdings $\times$ 15% = 60% < 100% (need ≥ 7 holdings for this mandate) 4 •
- holdings $\times$ 20% = 80% < 100% (need ≥ 5 holdings for this mandate) 4 •
- Min achievable β = 1.13 exceeds cap $\beta \leq 1.00$ •
- Sector 'Technology': 3/4 holdings $\rightarrow$ max total = 40% < 100% with current assets •
- Sector 'Technology': 3/4 holdings $\rightarrow$ max total = 50% < 100% with current assets •
- Sector 'Technology': 3/4 holdings $\rightarrow$ max total = 60% < 100% with current assets •

:To achieve full mandate compliance, add 💡

- Add 1+ new assets OR use Aggressive mandate (4 $\times$ 20% $\rightarrow$ 80% min) •
- Add 3+ new assets OR use Aggressive mandate (4 $\times$ 20% $\rightarrow$ 80% min) •
- Add 6+ new assets OR use Aggressive mandate (4 $\times$ 20% $\rightarrow$ 80% min) •
- Add diversifying sectors: XLV (Healthcare), TLT (Bonds), GLD (Gold), BRK-B, XLP •
- Add: TLT ($\beta \approx -0.5$), GLD ($\beta \approx 0.1$), USMV ($\beta \approx 0.7$), BRK-B ($\beta \approx 0.9$) •

Optimal Weight Comparison 📊

Aggressive 🚀	Balanced ⚖️	Conservative 🏛️	Current	Ticker
22.8%	1.0%	86.8%	35.3%	NVDA
64.2%	89.2%	1.0%	23.5%	AAPL
1.0%	1.0%	1.0%	23.5%	MSFT
12.0%	8.8%	11.2%	17.6%	TSLA
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0.80	0.54	1.13	0.85	Sharpe
1.50	1.20	2.30	1.70	Beta
4.24%-	4.39%-	5.10%-	3.93%-	CVaR/day
31.1%	31.4%	40.0%	30.6%	Ann. Vol
partial (relaxed: ⚠️ single-stock cap, sector cap)	partial ⚠️ (relaxed: single- stock cap, sector cap)	partial (relaxed: ⚠️ single-stock cap, beta cap, sector cap)	Baseline	Solver

Pension / Endowment | Solver:  partial (relaxed: single-stock cap, beta cap, sector cap)

Reason	on \$ \$100k	Δ	To	From	Ticker	Action
Underweight vs risk-optimal	\$51,459	51.5pp+	86.8%	35.3%	NVDA	↑ BUY 
Exceeds 10% mandate cap	\$22,529	22.5pp-	1.0%	23.5%	AAPL	SELL  ↓
Exceeds 10% mandate cap	\$22,529	22.5pp-	1.0%	23.5%	MSFT	SELL  ↓
Exceeds 10% mandate cap	\$6,400	6.4pp-	11.2%	17.6%	TSLA	SELL  ↓

Multi-Objective Trade-off Analysis

Status	Ann.Vol	Beta	CVaR/day	Sharpe	Strategy
Baseline	30.6%	1.70	3.93%-	0.85	Current portfolio
partial (relaxed:  single-stock cap, beta cap, sector cap)	40.0%	2.30 (0.60+)	5.10%- (1.17pp-)	1.13 (0.28+)	 Conservative
partial (relaxed:  single-stock cap, sector cap)	31.4%	1.20 (0.50-)	4.39%- (0.46pp-)	0.54 (0.31-)	Balanced 
partial (relaxed:  single-stock cap, sector cap)	31.1%	1.50 (0.20-)	4.24%- (0.31pp-)	0.80 (0.05-)	Aggressive 

Post-Rebalance Simulation

Best mandate:  Conservative · Pension / Endowment

Verdict	Δ	After	Before	Metric
Improved 	0.28+	1.13	0.85	Sharpe Ratio

Verdict	Δ	After	Before	Metric
Higher	9.35+	40.0%	30.6%	Ann. Volatility
Similar	0.60+	2.30	1.70	Portfolio Beta
Similar	1.17-	5.10%-	3.93%-	CVaR 95%/day
	—	Partial	Breach	Single-stock cap
	—	Near limit	Breach	Beta mandate

Portfolio Assessment

.CIO analysis temporarily unavailable

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Audit Trail

For compliance, reproducibility, and institutional trust

Value	Field
89b2dd13-c14a-4b0b-8fce-a7edae2562fe	Report ID
UTC 12:52:20 2026-03-30	Generated
2026-03-27	Price Data As-Of
(trading days 252) 2026-03-27 → 2025-03-28	Period Analysed
Yahoo Finance (yfinance) — 15-min delayed	Data Source
NVDA, AAPL, MSFT, TSLA	Tickers Fetched
(US 3-Month T-Bill) 4.50%	Risk-Free Rate
Historical Simulation · Brinson Attribution · Fama-French proxy	Methodology
CLARABEL convex QP (cvxpy) — strict constraint enforcement	Optimizer
sha256:0800c31e01776ba7...	Report Hash

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